

Anatomy of Cryptocurrency Perpetual Futures Returns

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Abstract

We analyse returns on cryptocurrency perpetual futures by first developing a cost-of-carry model tailored to digital assets. The model captures the link between spot and perpetual futures prices, implying a positive convenience yield and negligible off-chain storage costs. Furthermore, we employ a log-linear approximation to demonstrate that expected return of holding perpetual futures derive from the current log basis, the misperception of forward-looking spot price, and the expected futures-spot spreads over the “maturity” of the futures contract. We then assess a comprehensive set of 170 return predictors, classified into categories of basis, momentum, liquidity, size, and volatility. Sorting based on these predictors yields 63 statistically significant total returns (i.e. price movement plus funding fee yields, with each exceeding the 5 significance level). Finally, we demonstrate that a two-factor model, based on the log-basis and a price-volume relevant factor, effectively explains all 63 strategies, highlighting the role of systematic drivers in perpetual futures markets.

Keywords: Cryptocurrency, Perpetual Futures, Cost-of-Carry Model, Pricing Factors, Basis, Price-Volume

JEL code: C63, F47

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